01_eda

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1. Project Outline

We are working with the **UCI Online Retail** data set (id 352): roughly one year of transactions from a UK-based online gift retailer (Dec 2010 – Dec 2011). The project is unsupervised, aiming at customer segmentation.

Two cleaned tables produced by `data_prep.py` live in `../data/`:

- `online_retail_clean.csv` – transaction-level (one row per line item)
- `customer_features.csv` – customer-level RFM + behavioral extras

This notebook does the EDA only; modelling lives in subsequent files.

2. Load data

```r
tx <- read_csv("../data/online_retail_clean.csv",
               show_col_types =  FALSE)
customer <- read_csv("../data/customer_features.csv",
                     show_col_types = FALSE)

tx$InvoiceDate <- as_datetime(tx$InvoiceDate)
tx$Date <- as_date(tx$Date)

list(
  transactions = dim(tx),
  customers    = dim(customer),
  date_range   = range(tx$InvoiceDate),
  countries    = n_distinct(tx$Country)
)
```

```
## $transactions
## [1] 391150     14
##
## $customers
## [1] 4334   21
##
## $date_range
## [1] "2010-12-01 08:26:00 UTC" "2011-12-09 12:50:00 UTC"
```

```
##
## $countries
## [1] 37
```

```r
glimpse(tx)
```

```
## Rows: 391,150
## Columns: 14
## $ InvoiceNo   <dbl> 536365, 536365, 536365, 536365, 536365, 536365, 536365, 53~
## $ StockCode   <chr> "85123A", "71053", "84406B", "84029G", "84029E", "22752", ~
## $ Description <chr> "WHITE HANGING HEART T-LIGHT HOLDER", "WHITE METAL LANTERN~
## $ Quantity    <dbl> 6, 6, 8, 6, 6, 2, 6, 6, 6, 32, 6, 6, 8, 6, 6, 3, 2, 3, 3, ~
## $ InvoiceDate <dttm> 2010-12-01 08:26:00, 2010-12-01 08:26:00, 2010-12-01 08:2~
## $ UnitPrice   <dbl> 2.55, 3.39, 2.75, 3.39, 3.39, 7.65, 4.25, 1.85, 1.85, 1.69~
## $ CustomerID  <dbl> 17850, 17850, 17850, 17850, 17850, 17850, 17850, 17850, 17~
## $ Country     <chr> "United Kingdom", "United Kingdom", "United Kingdom", "Uni~
## $ Amount      <dbl> 15.30, 20.34, 22.00, 20.34, 20.34, 15.30, 25.50, 11.10, 11~
## $ Date        <date> 2010-12-01, 2010-12-01, 2010-12-01, 2010-12-01, 2010-12-0~
## $ Time        <time> 08:26:00, 08:26:00, 08:26:00, 08:26:00, 08:26:00, 08:26:0~
## $ Hour        <dbl> 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8~
## $ DOW         <chr> "Wednesday", "Wednesday", "Wednesday", "Wednesday", "Wedne~
## $ month       <dbl> 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12~
```

```r
glimpse(customer)
```

```
## Rows: 4,334
## Columns: 21
## $ CustomerID         <dbl> 12346, 12347, 12348, 12349, 12350, 12352, 12353, 12~
## $ Recency            <dbl> 326, 2, 75, 19, 310, 36, 204, 232, 214, 23, 33, 2, ~
## $ Frequency          <dbl> 1, 7, 4, 1, 1, 7, 1, 1, 1, 3, 1, 2, 4, 3, 1, 10, 2,~
## $ Monetary           <dbl> 77183.60, 4310.00, 1437.24, 1457.55, 294.40, 1385.7~
## $ TotalQuantity      <dbl> 74215, 2458, 2332, 630, 196, 526, 20, 530, 240, 157~
## $ UniqueProducts     <dbl> 1, 103, 21, 72, 16, 57, 4, 58, 13, 52, 131, 12, 214~
## $ AvgUnitPrice       <dbl> 1.040000, 2.644011, 0.692963, 4.237500, 1.581250, 4~
## $ FirstPurchase      <dttm> 2011-01-18 10:01:00, 2010-12-07 14:57:00, 2010-12--~
## $ LastPurchase       <dttm> 2011-01-18 10:01:00, 2011-12-07 15:52:00, 2011-09--~
## $ TenureDays         <dbl> 0, 365, 282, 0, 0, 260, 0, 0, 0, 302, 0, 149, 274, ~
## $ AvgOrderValue      <dbl> 77183.6000, 615.7143, 359.3100, 1457.5500, 294.4000~
## $ AvgOrderItems      <dbl> 74215.00000, 351.14286, 583.00000, 630.00000, 196.0~
## $ ReturnInvoices     <dbl> 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 2, 0, 0, 3, 0, ~
## $ Country            <chr> "United Kingdom", "Iceland", "Finland", "Italy", "N~
## $ ReturnRate         <dbl> 0.5000000, 0.0000000, 0.0000000, 0.0000000, 0.00000~
## $ log_Recency        <dbl> 5.789960, 1.098612, 4.330733, 2.995732, 5.739793, 3~
## $ log_Frequency      <dbl> 0.6931472, 2.0794415, 1.6094379, 0.6931472, 0.69314~
## $ log_Monetary       <dbl> 11.253955, 8.368925, 7.271175, 7.285198, 5.688330, ~
## $ log_TotalQuantity  <dbl> 11.214735, 7.807510, 7.754910, 6.447306, 5.283204, ~
## $ log_UniqueProducts <dbl> 0.6931472, 4.6443909, 3.0910425, 4.2904594, 2.83321~
## $ log_AvgOrderValue  <dbl> 11.253955, 6.424406, 5.886965, 7.285198, 5.688330, ~
```

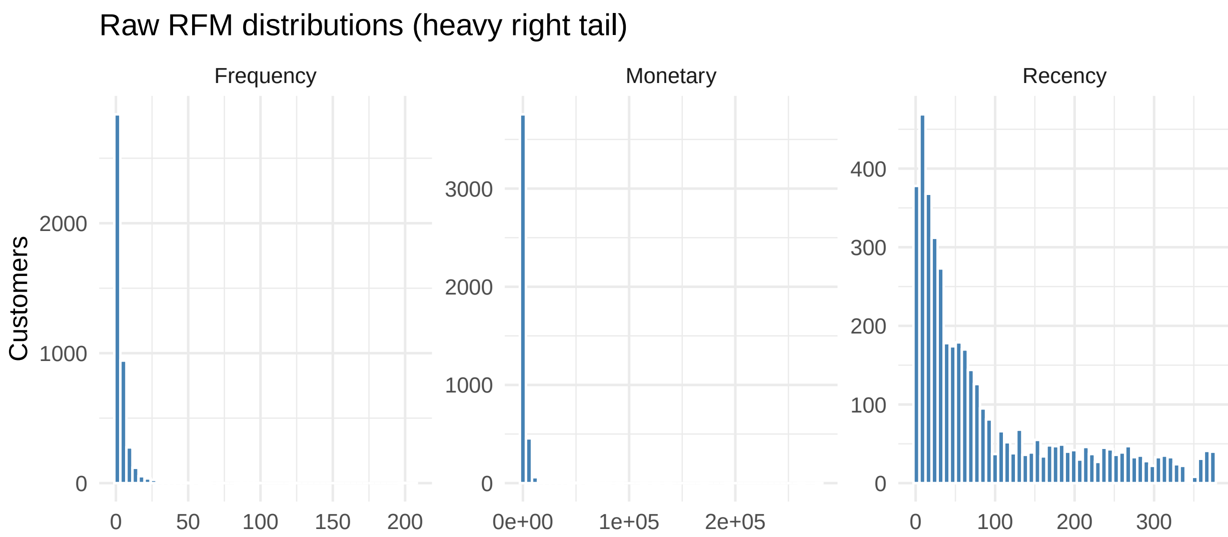
3. Headline numbers

```
tibble(
  customer = n_distinct(tx$CustomerID),
  invoices = n_distinct(tx$InvoiceNo),
  unique_products = n_distinct(tx$StockCode),
  line_items = nrow(tx),
  total_revenue = sum(tx$Amount),
  avg_basket = mean(tx$Amount)
) |>
  pivot_longer(everything(), names_to = "metric", values_to = "value") |>
  mutate(value = comma(value, accuracy = 0.01))
```

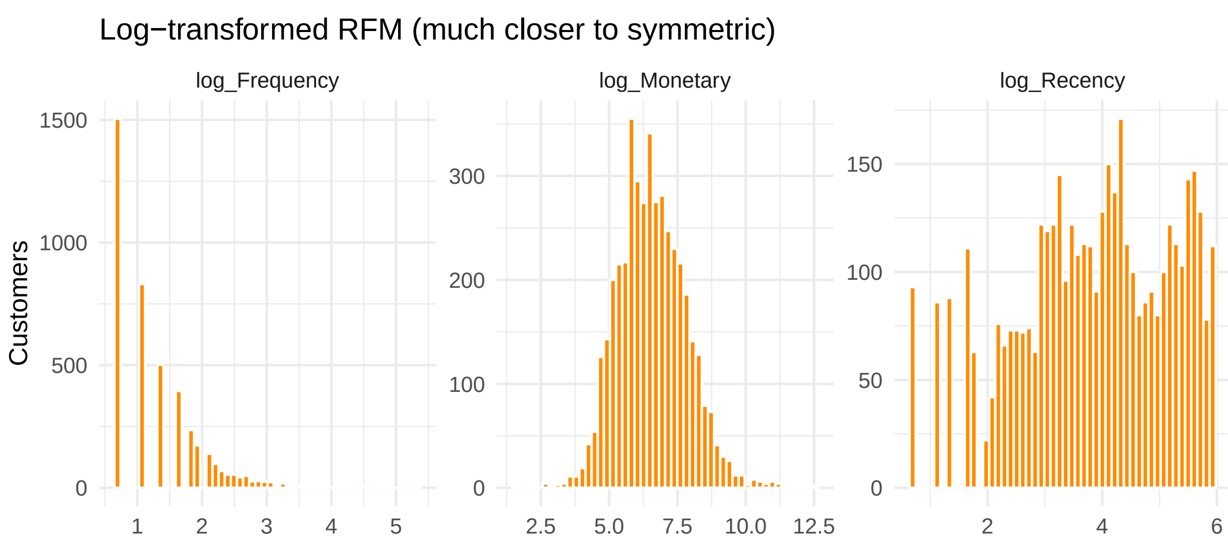
```
## # A tibble: 6 x 2
##   metric          value
##   <chr>           <chr>
## 1 customer        4,334.00
## 2 invoices        18,402.00
## 3 unique_products 3,659.00
## 4 line_items      391,150.00
## 5 total_revenue   8,737,227.64
## 6 avg_basket      22.34
```

4. Distribution of RFM features (raw vs. log)

```
customer |>
  select(Recency, Frequency, Monetary) |>
  pivot_longer(everything(), names_to = "feature", values_to = "value") |>
  ggplot(aes(value)) +
  geom_histogram(bins = 50, fill = "steelblue", colour = "white") +
  facet_wrap(~ feature, scales = "free", nrow = 1) +
  labs(title = "Raw RFM distributions (heavy right tail)",
       x = NULL, y = "Customers")
```



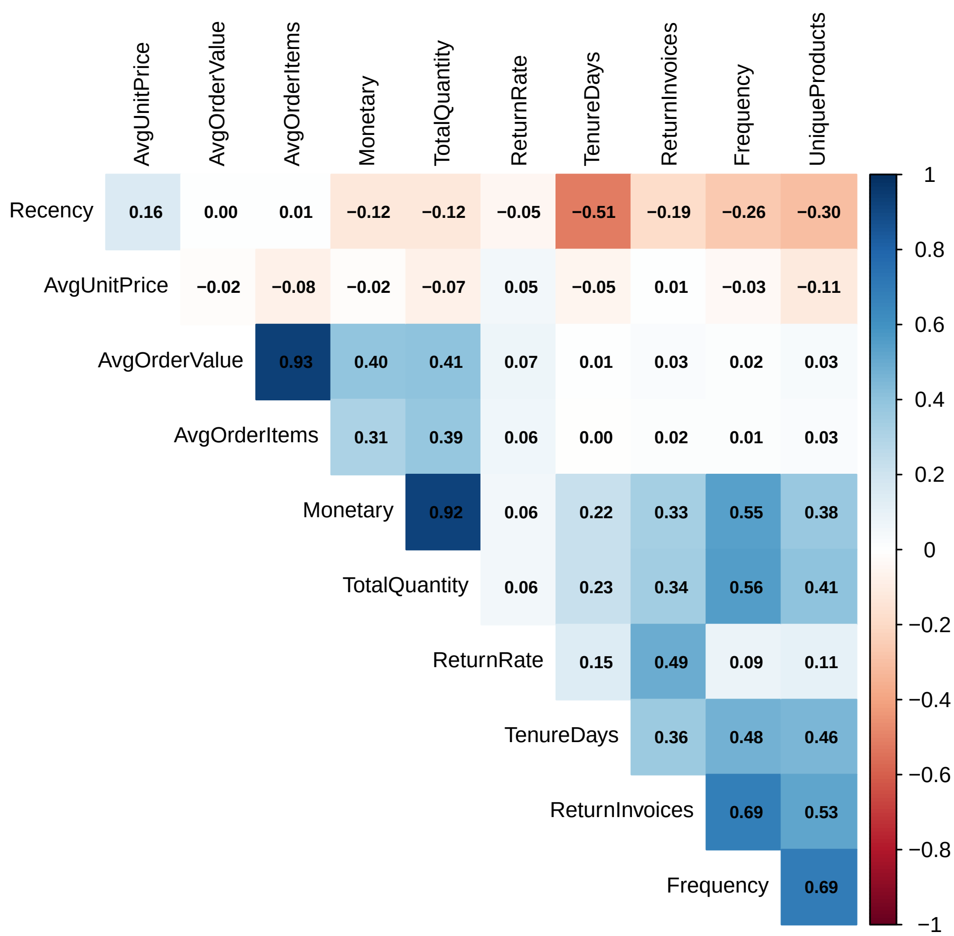

```r
customer |>
  select(log_Recency, log_Frequency, log_Monetary) |>
  pivot_longer(everything(), names_to = "feature", values_to = "value") |>
  ggplot(aes(value)) +
  geom_histogram(bins = 50, fill = "darkorange", colour = "white") +
  facet_wrap(~ feature, scales = "free", nrow = 1) +
  labs(title = "Log-transformed RFM (much closer to symmetric)",
       x = NULL, y = "Customers")
```



The log scale also makes the wholesale tail in `Monetary` legible -under raw scaling those customers are near-invisible, on log scale they form a distinct upper shoulder of the distribution.

5. Corraltion structure

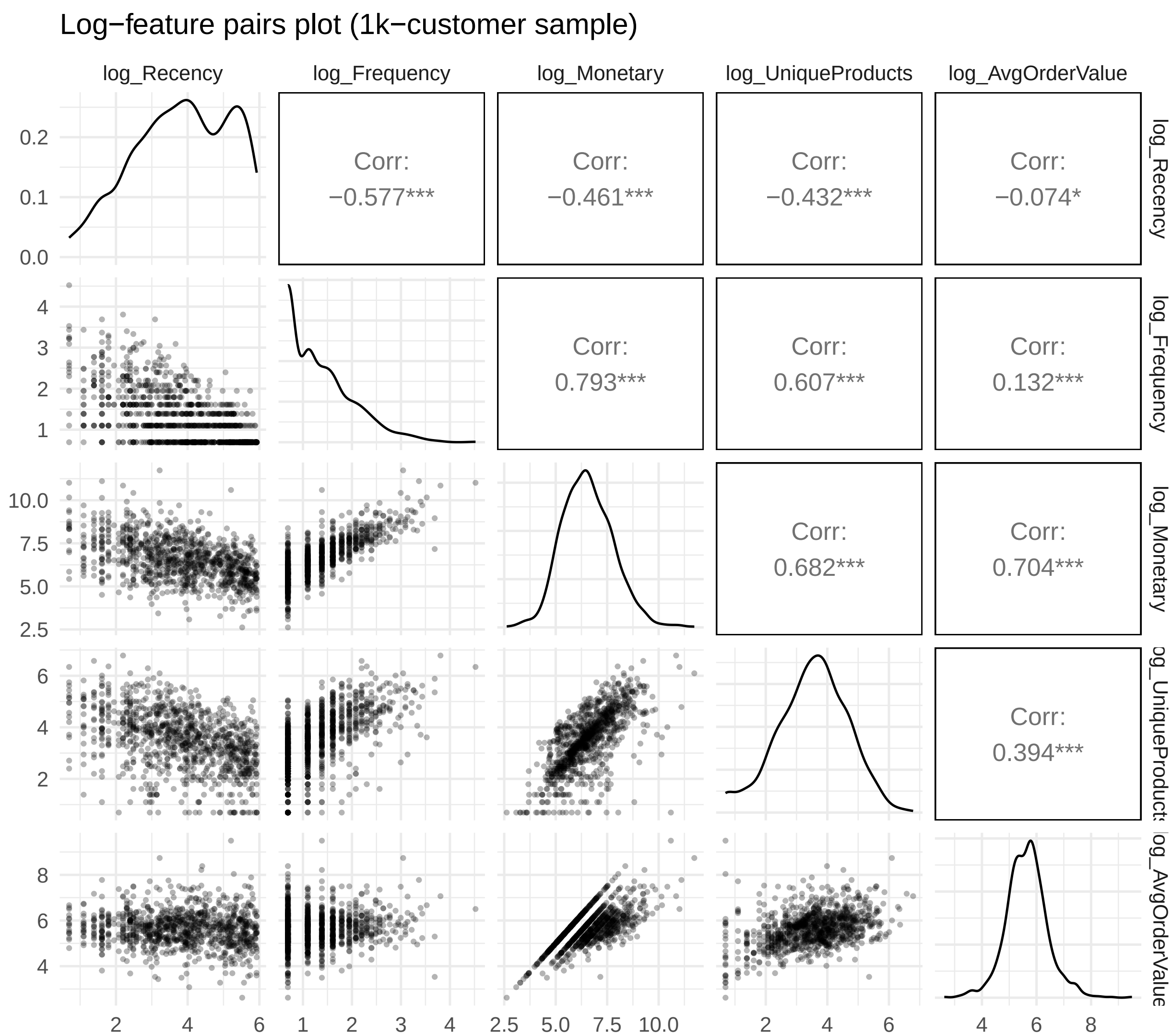
```r
num_cols <- customer |>
  select(Recency, Frequency, Monetary,
         TotalQuantity, UniqueProducts,
         AvgUnitPrice,
         AvgOrderValue, AvgOrderItems,
         TenureDays, ReturnInvoices, ReturnRate)

corrplot::corrplot(
  cor(num_cols, use = "pairwise.complete.obs"),
  method = "color", type = "upper", order = "hclust",
  addCoef.col = "black", number.cex = 0.7,
  tl.col = "black", tl.cex = 0.8, diag = FALSE
)
```



* `Monetary`, `TotalQuantity`, and `Frequency` are tightly correlated — expected, since spending more usually means buying more in more orders. * `AvgOrderValue` and `AvgBasketValue` carry similar signal; we'll likely keep only one in the clustering input. * `Recency` is largely independent of the spending features — it's its own axis of variation. That's why R, F, M is the canonical triple.

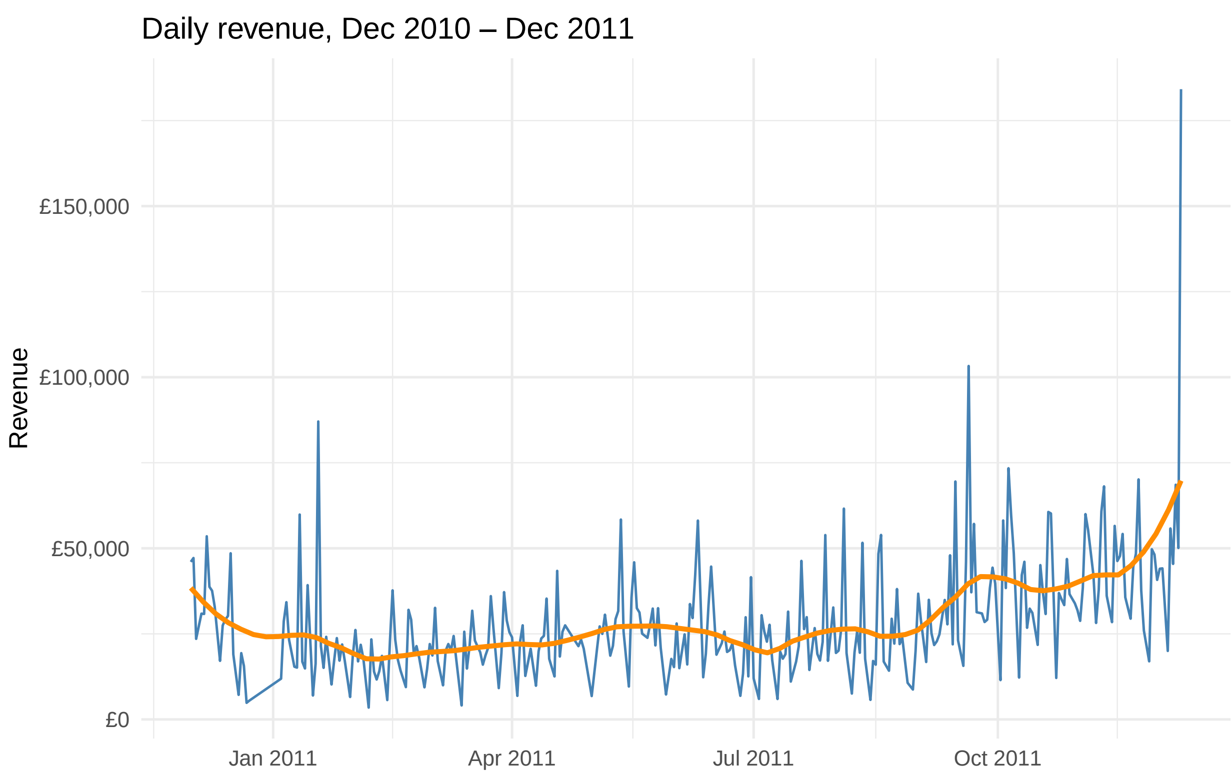
6. Pairs plot on the modelling set

```r
set.seed(1)
customer |>
  slice_sample(n = 1000) |>
  select(log_Recency, log_Frequency, log_Monetary,
         log_UniqueProducts, log_AvgOrderValue) |>
  GGally::ggpairs(progress = FALSE,
                  lower = list(continuous = wrap("points", alpha = 0.3,
                                                  size = 0.6))) +
  labs(title = "Log-feature pairs plot (1k-customer sample)")
```



7. Sales over time

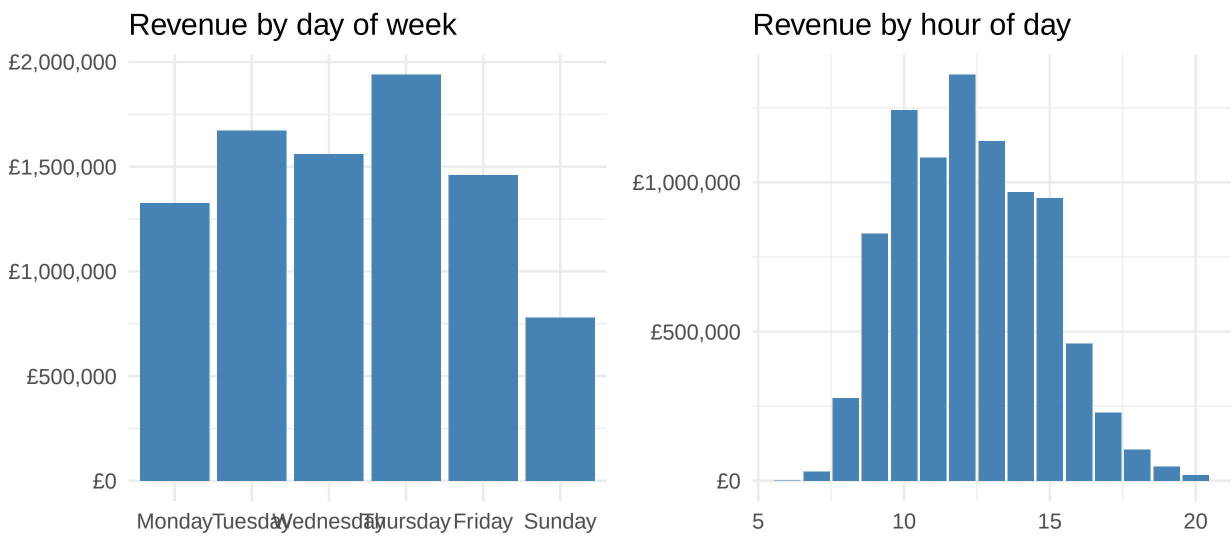
```r
daily <- tx |>
  group_by(Date) |>
  summarise(revenue = sum(Amount),
            invoices = n_distinct(InvoiceNo), .groups = "drop")

ggplot(daily, aes(Date, revenue)) +
  geom_line(colour = "steelblue") +
  geom_smooth(
    method = "loess",
    span = 0.2,
    se = FALSE,
    colour = "darkorange"
  ) +
  scale_y_continuous(labels = label_comma(prefix = "\u00A3")) +
  labs(
    title = "Daily revenue, Dec 2010 \u2013 Dec 2011",
    x = NULL,
    y = "Revenue"
  )
```



The Q4 2011 ramp is the holiday gift season — exactly what we'd expect for this retailer. Good sanity check that the data makes sense.

```r
library(gridExtra)

pound <- "\u00A3"

p_dow <- tx |>
```

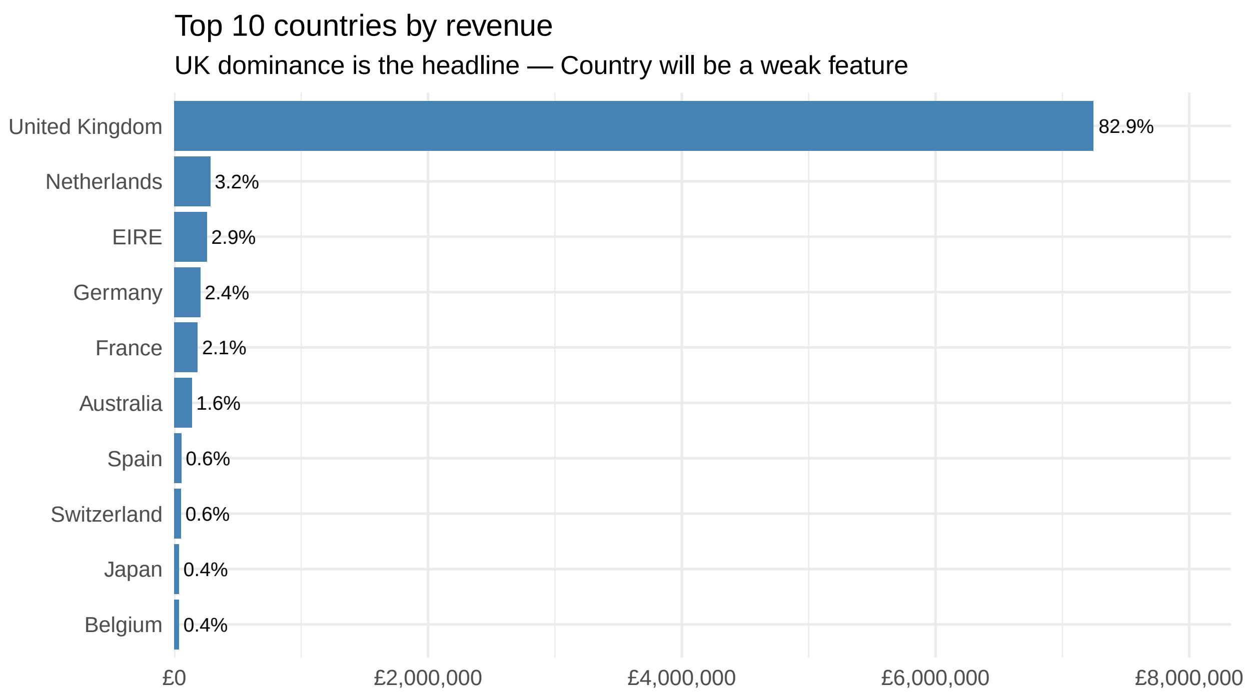
```
  count(
    DOW = factor(
      DOW,
      levels = c(
        "Monday", "Tuesday", "Wednesday",
        "Thursday", "Friday", "Saturday", "Sunday"
      )
    ),
    wt = Amount,
    name = "revenue"
  ) |>
  ggplot(aes(DOW, revenue)) +
  geom_col(fill = "steelblue") +
  scale_y_continuous(labels = label_comma(prefix = pound)) +
  labs(title = "Revenue by day of week", x = NULL, y = NULL)

p_hour <- tx |>
  count(Hour, wt = Amount, name = "revenue") |>
  ggplot(aes(Hour, revenue)) +
  geom_col(fill = "steelblue") +
  scale_y_continuous(labels = label_comma(prefix = pound)) +
  labs(title = "Revenue by hour of day", x = NULL, y = NULL)

gridExtra::grid.arrange(p_dow, p_hour, nrow = 1)
```



8. Geography

```
pound <- "\u00A3"
emdash <- "\u2014"

tx |>
  count(Country, wt = Amount, name = "revenue", sort = TRUE) |>
```

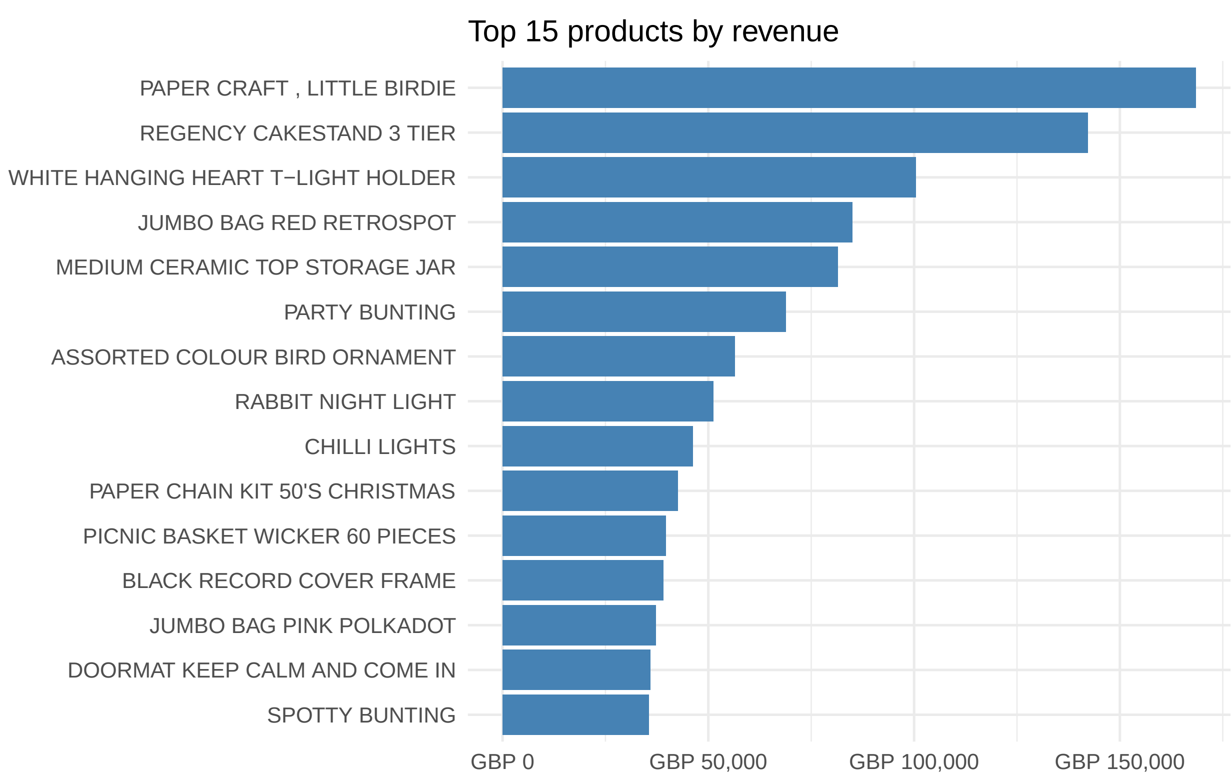
```r
  mutate(
    share = revenue / sum(revenue),
    Country = fct_reorder(Country, revenue)
  ) |>
  slice_max(revenue, n = 10) |>
  ggplot(aes(revenue, Country)) +
  geom_col(fill = "steelblue") +
  geom_text(
    aes(label = percent(share, accuracy = 0.1)),
    hjust = -0.1,
    size = 3
  ) +
  scale_x_continuous(
    labels = label_comma(prefix = pound),
    expand = expansion(mult = c(0, 0.15))
  ) +
  labs(
    title = "Top 10 countries by revenue",
    subtitle = paste("UK dominance is the headline", emdash,
                     "Country will be a weak feature"),
    x = NULL,
    y = NULL
  )
```



9. Top products

```r
tx |>
  mutate(
```

```r
    Description = iconv(Description, from = "", to = "UTF-8", sub = "")
  ) |>
  group_by(StockCode, Description) |>
  summarise(
    revenue = sum(Amount, na.rm = TRUE),
    qty     = sum(Quantity, na.rm = TRUE),
    .groups = "drop"
  ) |>
  slice_max(revenue, n = 15) |>
  mutate(Description = fct_reorder(Description, revenue)) |>
  ggplot(aes(revenue, Description)) +
  geom_col(fill = "steelblue") +
  scale_x_continuous(labels = label_comma(prefix = "GBP ")) +
  labs(
    title = "Top 15 products by revenue",
    x = NULL,
    y = NULL
  )
```

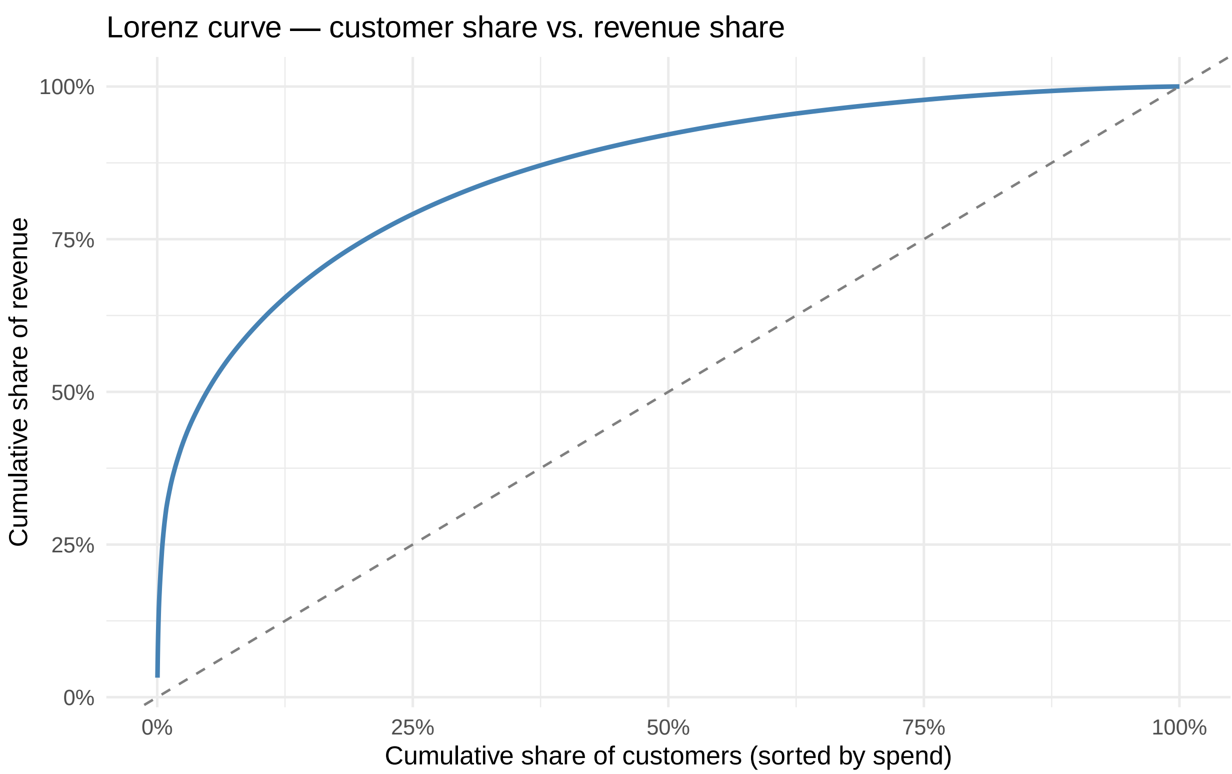


10. Revenue concentration (Pareto check)

How concentrated is revenue across customers? This frames the outlier question for clustering.

```r
pareto <- customer |>
  arrange(desc(Monetary)) |>
  mutate(rank        = row_number(),
```

```r
         pct_cust     = rank / n(),
         cum_revenue = cumsum(Monetary) / sum(Monetary))

ggplot(pareto, aes(pct_cust, cum_revenue)) +
  geom_line(colour = "steelblue", linewidth = 1) +
  geom_abline(linetype = "dashed", colour = "grey50") +
  scale_x_continuous(labels = percent_format()) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Lorenz curve - customer share vs. revenue share",
       x = "Cumulative share of customers (sorted by spend)",
       y = "Cumulative share of revenue")
```



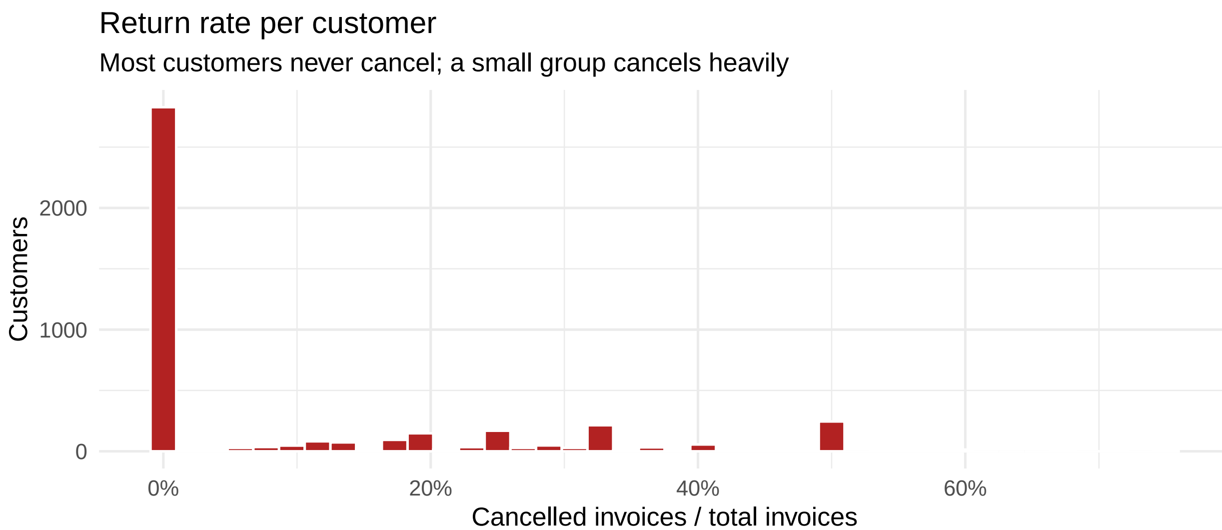

```r
pareto |>
  filter(pct_cust %in% c(0.01, 0.05, 0.10, 0.20, 0.50)) |>
  transmute(top_pct_customers = percent(pct_cust),
            share_of_revenue  = percent(cum_revenue, accuracy = 0.1))
```

```
## # A tibble: 1 x 2
##   top_pct_customers share_of_revenue
##   <chr>             <chr>
## 1 50%               92.2%
```

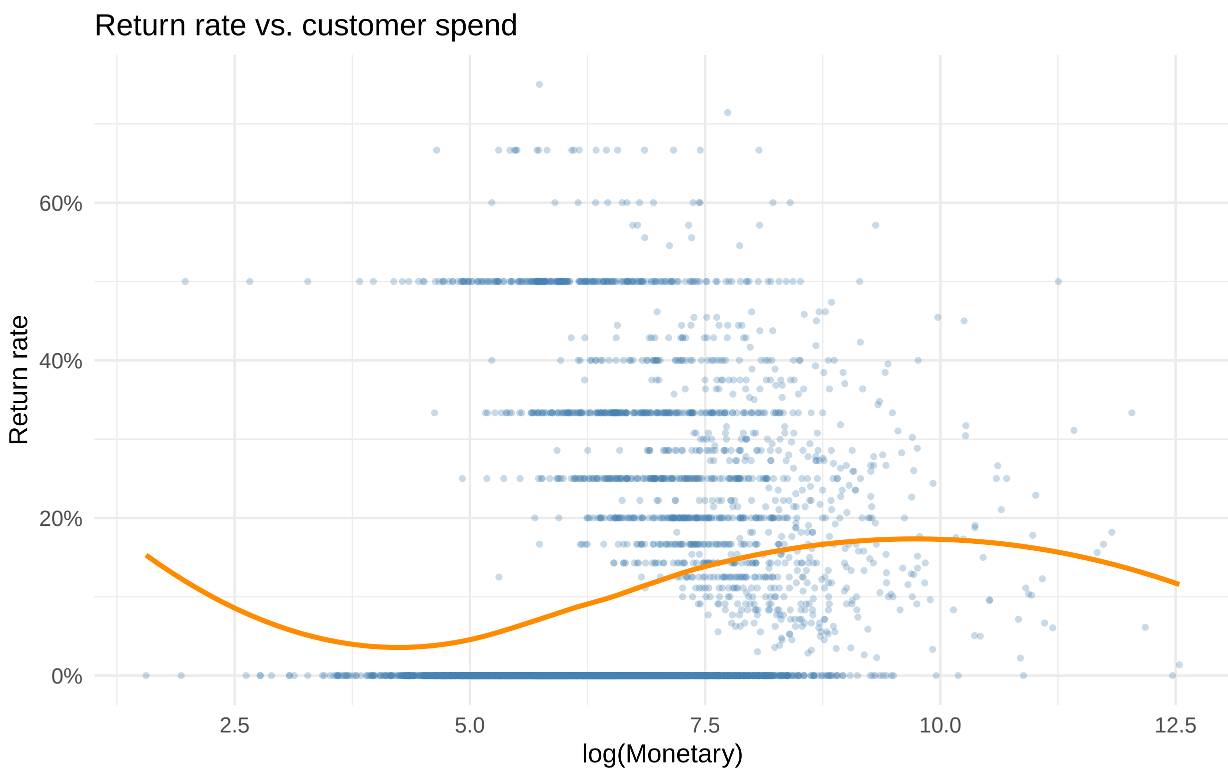
If a tiny share of customers controls a large share of revenue, they'll dominate any unscaled clustering. This confirms the plan to cluster on log-features rather than raw.

11. Returns / cancellations

```
customer |>
  ggplot(aes(ReturnRate)) +
  geom_histogram(bins = 40, fill = "firebrick", colour = "white") +
  scale_x_continuous(labels = percent_format()) +
  labs(title = "Return rate per customer",
       subtitle = "Most customers never cancel; a small group cancels heavily",
       x = "Cancelled invoices / total invoices", y = "Customers")
```




```
ggplot(customer, aes(log_Monetary, ReturnRate)) +
  geom_point(alpha = 0.3, size = 0.8, colour = "steelblue") +
  geom_smooth(method = "loess", se = FALSE, colour = "darkorange") +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Return rate vs. customer spend",
       x = "log(Monetary)", y = "Return rate")
```


Return rate vs. customer spend
Return rate
60%
40%
20%
0%
2.5
5.0
7.5
10.0
12.5
log(Monetary)